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Lindblad Master Equation:
Generator of Markovian CPTP Maps

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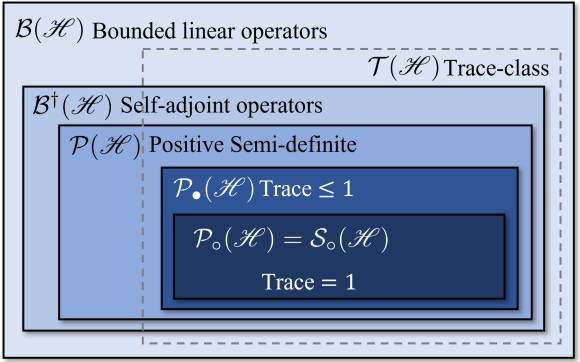
Let us start our discussion with closed quantum system with the following notations:

- Hilbert space: $\mathscr{H}$;
- Bounded linear operators: $A \in \mathcal{B}(\mathscr{H}_1, \mathscr{H}_2) : \mathscr{H}_1 \to \mathscr{H}_2$, and $\mathcal{B}(\mathscr{H}) \equiv \mathcal{B}(\mathscr{H}, \mathscr{H})$;
- Hermitian operator: $\mathcal{H} \in \mathcal{B}^\dagger(\mathscr{H})$ (self-adjoint)
- Trace-class operators: $\mathcal{T}(\mathscr{H}) = \left\{ M \in \mathcal{B}(\mathscr{H}) \,\middle|\, \mathrm{Tr}\left\{\sqrt{M^\dagger M}\right\} < \infty \right\}$, with:

$$\langle \sigma, A \rangle = \mathrm{Tr}\left\{\sigma^\dagger A\right\} \qquad (\sigma \in \mathcal{T}(\mathscr{H}), A \in \mathcal{B}(\mathscr{H})). \tag{1}$$

- Quantum state (density operator): $\rho \in \mathcal{S}_\circ(\mathscr{H}) \equiv \mathcal{B}_1(\mathscr{H}) \subset \mathcal{T}(\mathscr{H})$, with Born rule

$$\langle A \rangle = \langle \rho, A \rangle = \mathrm{Tr}\left\{\rho A\right\}. \tag{2}$$



Note that in finite dimensional $\mathscr{H}$, $\mathcal{T}(\mathscr{H}) = \mathcal{B}(\mathscr{H})$.

A quantum system $(\mathscr{H}, \rho, \mathcal{H})$ evolves according to the Liouville-von Neumann equation (in the Schrödinger picture):

Liouvillian, $\mathcal{L} = -i\mathrm{ad}_{\mathcal{H}}$

$$\dot{\rho} = -i[\mathcal{H}, \rho] \equiv \mathcal{L}\,\rho. \tag{3}$$

The evolution of a closed quantum system is trivial. For $\mathcal{H}$ being time-independent, the state evolves unitarily as

$$\begin{aligned} \dot{\rho} = -i[\mathcal{H}, \rho] \;\longleftrightarrow\; \rho(t) = \mathrm{e}^{-i\mathcal{H}(t-t_0)}\rho(t_0)\mathrm{e}^{i\mathcal{H}(t-t_0)}, \\ \dot{\rho} = \mathcal{L}\rho \;\longleftrightarrow\; \rho(t) = \mathrm{e}^{\mathcal{L}(t-t_0)}\rho(t_0). \end{aligned} \tag{4}$$

For $\mathcal{H} = \mathcal{H}(t)$, we have

$$\rho(t) = V_{t,t_0} \cdot \rho(t_0) \qquad (t \geq t_0), \tag{5}$$

where the propagator V_{t,t_0} is defined as

$$V_{t,t_0} = T_{\leftarrow}\left\{\exp\left(\int_{t_0}^{t} \mathcal{L}(\tau)\,\mathrm{d}\tau\right)\right\} \overset{\text{discretize}}{\approx} \mathrm{e}^{\mathcal{L}(t_{N-1})\,\mathrm{d}\tau} \cdots \mathrm{e}^{\mathcal{L}(t_0)\,\mathrm{d}\tau}, \tag{6}$$

with $T_{\leftarrow}$ being the time-ordering operator (from past to future in the $\leftarrow$ direction).

However, for *open quantum systems*, the system we are interested in will *interact* with the surrounding environment. Such interactions create non-unitary evolutions and irreversible effects on the system.

To make the analysis of our system tractable (the environment may have infinite degrees of freedom), we require a *master equation* to describe the evolution of our quantum state. This is the celebrated GKSL equation[1]:

Hamiltonian, $\mathcal{H} \in \mathcal{B}^\dagger(\mathscr{H})$

anti-commutator, $\{A, B\} = AB + BA$

$$\dot{\rho}(t) = \mathcal{L}\,\rho(t) := -i[\mathcal{H}, \rho(t)] + \sum_{\alpha \geq 1} \gamma_\alpha \left(L_\alpha \rho(t) L_\alpha^\dagger - \frac{1}{2}\{ L_\alpha^\dagger L_\alpha, \rho(t) \} \right). \quad (7)$$

Lindbladian

≥ 0

Lindblad operators, $L_\alpha \in \mathcal{B}(\mathscr{H})$

[1] Named after Vittorio Gorini, Andrzej Kossakowski, George Sudarshan and Göran Lindblad.

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Let us consider in the Schrödinger's picture, where a quantum system $\Phi : \mathcal{B}(\mathscr{H}_1) \to \mathcal{B}(\mathscr{H}_2)$ transforms the state ρ into ρ'. What should Φ satisfy?

1. **Convex-Linearity**: For arbitrary density operators ρ_1, ρ_2 and probability $0 \leq \lambda \leq 1$, we require that

$$\Phi\left(\lambda\rho_1 + (1-\lambda)\rho_2\right) = \lambda\Phi(\rho_1) + (1-\lambda)\Phi(\rho_2). \tag{8}$$

2. **Trace Preserving (TP)**: We require that the total probability to conserve[2],

$$\mathrm{Tr}\left\{\Phi(\rho)\right\} = \mathrm{Tr}\left\{\rho\right\}. \tag{9}$$

3. **Positivity**: Probability should always be non-negative.

$$\rho \succeq 0 \Rightarrow \Phi(\rho) \succeq 0. \tag{10}$$

[2] For the cases where the probability doesn't conserve, it involves particle generation or annihilation.

Positivity is not enough!

Consider our system as $(\mathscr{H}, \rho, \Phi)$. If we include an ancillary system $(\mathscr{H}_E, \rho_E, \mathbb{1}_E)$, then the total evolution on the Hilbert space $\mathscr{H} \otimes \mathscr{H}_E$ for the joint density operator $\rho \otimes \rho_E$ will be

$$\rho \otimes \rho_E \longmapsto (\Phi \otimes \mathbb{1}_E)(\rho \otimes \rho_E),$$

this should remain as a *valid* density operator.

Complete Positivity

A positive map $\Phi : \mathcal{B}(\mathscr{H}_1) \to \mathcal{B}(\mathscr{H}_2)$ is said to be k-positive if

$$\Phi \otimes \mathbb{1}_k \tag{11}$$

is positive. It is further called completely positive (CP) if it is k-positive for all $k \in \mathbb{N}$.

Example. (Positivity is NOT enough)

Consider the positive map of transposition $(\cdot)^\mathsf{T} : \rho \mapsto \rho^\mathsf{T}$ acting on $\mathscr{H}_1 \cong \mathbb{C}^2$. Consider an ancillary system of $\mathscr{H}_2 \cong \mathbb{C}^2$. Then the map $(\cdot)^\mathsf{T} \otimes \mathbb{1}_2$ is not positive when acting on

$$\mathscr{H}_1 \otimes \mathscr{H}_2 \ni \rho = \frac{1}{2}\left(\begin{bmatrix}1 & 0\\0 & 0\end{bmatrix} \otimes \begin{bmatrix}0 & 0\\0 & 1\end{bmatrix} + \begin{bmatrix}0 & 1\\0 & 0\end{bmatrix} \otimes \begin{bmatrix}0 & 0\\1 & 0\end{bmatrix} + \begin{bmatrix}0 & 0\\1 & 0\end{bmatrix} \otimes \begin{bmatrix}0 & 1\\0 & 0\end{bmatrix} + \begin{bmatrix}0 & 0\\0 & 1\end{bmatrix} \otimes \begin{bmatrix}1 & 0\\0 & 0\end{bmatrix}\right).$$

$$\left((\cdot)^\mathsf{T} \otimes \mathbb{1}_2\right)\rho = \frac{1}{2}\begin{bmatrix}0 & 0 & 0 & 1\\0 & 1 & 0 & 0\\0 & 0 & 1 & 0\\1 & 0 & 0 & 0\end{bmatrix} \text{ is indefinite.}$$

Hence we have concluded that:

Quantum Channel

For a map $\Phi : \mathcal{B}(\mathscr{H}_1) \to \mathcal{B}(\mathscr{H}_2)$ to be a valid *quantum channel* / *quantum map*, it has to be

- convex-linear,
- completely positive and trace-preserving (CPTP).

Adjoint Map

Given map $\Phi : \mathcal{B}(\mathscr{H}_1) \to \mathcal{B}(\mathscr{H}_2)$, the adjoint map is $\Phi^\dagger : \mathcal{B}(\mathscr{H}_2) \to \mathcal{B}(\mathscr{H}_1)$ satisfying

$$\langle \Phi(\rho), \sigma \rangle_2 \equiv \mathrm{Tr}_2 \left\{ \Phi(\rho) \cdot \sigma \right\} = \mathrm{Tr}_1 \left\{ \rho \cdot \Phi^\dagger(\sigma) \right\} \equiv \langle \rho, \Phi^\dagger(\sigma) \rangle_1 \tag{12}$$

for all $\rho \in \mathcal{B}(\mathscr{H}_1)$ and $\sigma \in \mathcal{B}(\mathscr{H}_2)$.

Theorems on Adjoint Map

A map Φ is:

- CP iff $\Phi^\dagger$ is CP;
- TP iff $\Phi^\dagger$ is unital, i.e. $\Phi^\dagger(\mathbb{1}_2) = \mathbb{1}_1$.

For a detailed description and proofs on quantum channel, please refer to [Led23].

Next, let us examine the notion of Markovianity of a quantum channel [WC08].

Markovian Channel

A channel Φ_t is called Markovian if it is an element of a *CP continuous one-parameter semigroup*. That is, there exists a generator $\mathcal{L} : \mathcal{B}(\mathscr{H}) \to \mathcal{B}(\mathscr{H})$ with $\mathcal{L}^\dagger(\mathbb{1}) = 0$, such that

$$\Phi_t = \mathrm{e}^{t\mathcal{L}} \tag{13}$$

is CPTP for all $t \geq 0$.

For our case of the GKSL equation $\dot{\rho} = \mathcal{L}\rho$, in which $\mathcal{L}$ is time-independent and $\mathcal{L}^\dagger(\mathbb{1}) = \mathbb{1}$, the channel is automatically Markovian and CPTP by definition.

For the cases where $\mathcal{L}$ is time-dependent, we call the system *time-dependent Markovian*. The channel can be described by the following quantum dynamical semigroup:

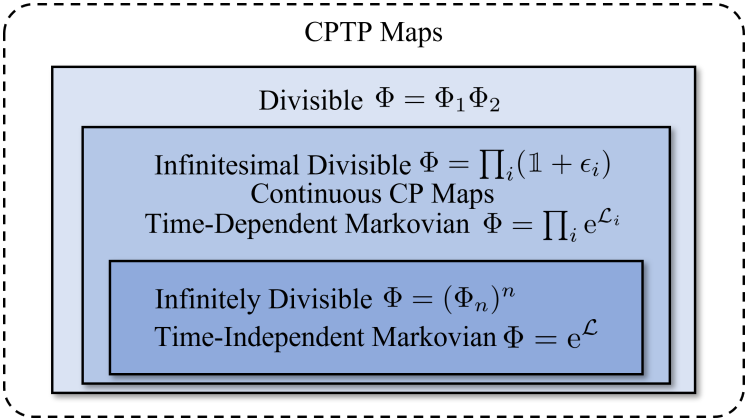
$$\Phi_t = T_{\leftarrow}\left\{\exp\left(\int_0^t \mathcal{L}\,\mathrm{d}\tau\right)\right\}. \tag{14}$$

In fact, it is of the form

$$\Phi_t = \prod_i \exp(\mathcal{L}_i\,\mathrm{d}\tau) = \prod_i (\mathbb{1} + \mathcal{L}_i\,\mathrm{d}\tau). \tag{15}$$

We term this kind of channel *infinitesimal divisible*.

We can summarize divisibility as below[3]: for Lindblad operator $\mathcal{L}$,



[3] This figure is adapted from Fig. 1 in [WC08].

[WC08] Lemma 1 (Markovian Approximation)

For every quantum channel $\Phi : \mathcal{B}(\mathscr{H}) \to \mathcal{B}(\mathscr{H})$, $\{\mathrm{e}^{t(\Phi - \mathbb{1})}\}$ is a completely positive semigroup.

The proof requires the notion of Kraus operators that will be discussed later. By showing that the generator $\Phi - \mathbb{1}$ is of the form of GKSL equation, the lemma is proven. ■

It is also shown in [WC08] that, amazingly, all kinds of channels, whether they be indivisible, divisible, not infinitesimal divisible, Markovian, can be found in the vicinity of the identity.

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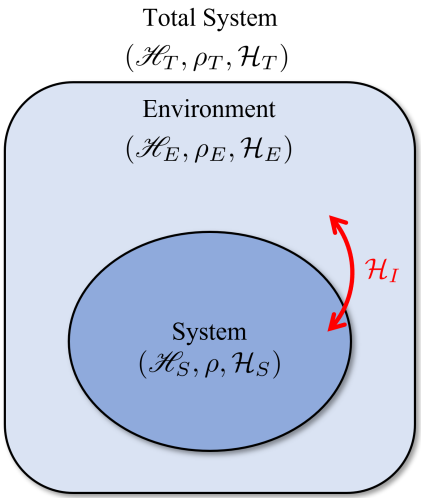
The combined total system of $\mathscr{H}_T = \mathscr{H}_S \otimes \mathscr{H}_E$ can be seen as evolving via the joint Hamiltonian

$$\mathcal{H}_T = \mathcal{H}_S \otimes \mathbb{1}_E + \mathbb{1}_S \otimes \mathcal{H}_E + \underbrace{\mathcal{H}_I}_{\text{interaction}}. \tag{16}$$

Hence the total system at time t will evolve (in the Schrödinger's picture) as

$$\dot{\rho}_T(t) = -i[\mathcal{H}_T, \rho_T(t)]$$
$$\rightarrow \rho_T(t) = U_{t,0} \cdot \rho_T(0) \cdot U_{t,0}^{\dagger}.$$

The interaction involved introduces noises, dissipation, and irreversibility to the system $\rho(t)$.



For the total state, it evolves unitarily as

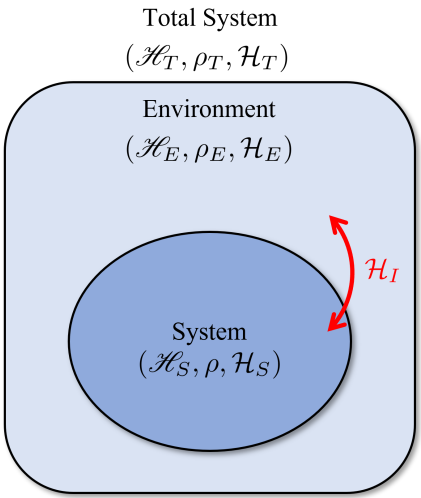
$$\rho_T(t) = U_{t,0} \cdot \rho_T(0) \cdot U_{t,0}^\dagger,$$

with the unitary propagator of the form

$$U_{t,0} = T_{\leftarrow}\left\{\exp\left(\int_0^t \mathcal{H}_T \,\mathrm{d}t\right)\right\}. \tag{17}$$

The evolution to the system ρ will simply be obtained by partial trace:

$$\rho(t) = \mathrm{Tr}_E\left\{\rho_T(t)\right\}. \tag{18}$$



The idea of representing a channel by purification is characterized as below:

Stinespring's Dilation

For any quantum channel $\Phi : \mathcal{B}(\mathscr{H}_1) \to \mathcal{B}(\mathscr{H}_2)$, there exists a Hilbert space $\mathscr{H}_E$, a pure state $\rho_E \in \mathcal{S}_\circ(\mathscr{H}_E)$, and an isometry $U \in \mathcal{B}(\mathscr{H}_1 \otimes \mathscr{H}_E, \mathscr{H}_2 \otimes \mathscr{H}_E)$, such that for all $\rho \in \mathcal{S}_\circ(\mathscr{H}_1)$,

$$\Phi(\rho) = \mathrm{Tr}_E \left\{ U(\rho \otimes \rho_E)U^\dagger \right\}. \tag{19}$$

Note that this is a reduced version of the full theorem.

An isometry $U \in \mathcal{B}(\mathscr{H}_1, \mathscr{H}_2)$ preserves the inner product: for all $v, w \in \mathscr{H}_1$, $\langle Uv, Uw \rangle_2 = \langle v, w \rangle_1$. This holds for $U^\dagger U = \mathbb{1}_1$. For $\mathscr{H}_1 = \mathscr{H}_2$, an isometry is simply a unitary operator.

$$\begin{tikzcd}
\rho_T(t_0) = \rho(t_0) \otimes \rho_E \arrow[r, "\text{unitary evolution}"] \arrow[d, "\mathrm{Tr}_E"'] & \rho_T(t) = U_{t,t_0} \cdot \rho(t_0) \cdot U_{t,t_0}^\dagger \arrow[d, "\mathrm{Tr}_E"] \\
\rho(t_0) \arrow[r, "\text{dynamical map}"] & \rho(t) = V_{t,t_0} \cdot \rho(t_0)
\end{tikzcd}$$

The reduced system ρ evolves according to the *dynamical map* [BP07, pp. 109–119]

$$V_{t,t_0} : \mathcal{S}_\circ(\mathscr{H}) \to \mathcal{S}_\circ(\mathscr{H}) \qquad (t \geq t_0), \tag{20}$$

such that

1. $V_{t,t} = \mathbb{1}$,
2. $V_{t,s} \cdot V_{s,t_0} = V_{t,t_0}$ $(t \geq s \geq t_0)$.

This forms a dynamical semigroup (a group without inverse). For a Hamiltonian (non-dissipative) system, the semigroup can be extended to a group.

By setting $V_{t,0} \equiv V(t)$, we obtain $\rho(t) = V(t)\rho(0)$. Then we have its law of motion / master equation as

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(V(t)\rho(0)\right) = \mathcal{L}\left(V(t)\rho(0)\right),$$

or simply as

$$\frac{\mathrm{d}}{\mathrm{d}t}V(t) = \mathcal{L} \cdot V(t). \tag{21}$$

We call the Lindbladian $\mathcal{L}$ the infinitesimal generator of the quantum dynamical semigroup $V(t)$. Our job is to find a way to characterize the infinitesimal generator for an open quantum system!

Remember that for the GKSL equation, we have the infinitesimal generator of it as

$$\begin{aligned}\dot{\rho}(t) &= \mathcal{L}\rho(t) \\ &= -i\,[\mathcal{H}, \rho(t)] + \sum_{\alpha \geq 1} \gamma_\alpha \left(L_\alpha \rho(t) L_\alpha^\dagger - \frac{1}{2}\{L_\alpha^\dagger L_\alpha, \rho(t)\} \right) \\ &= \underbrace{-i[\mathcal{H}, \rho(t)]}_{\text{Hamiltonian part}} + \underbrace{\mathcal{D}\{\rho(t)\}}_{\text{dissipative part}} \quad .\end{aligned}$$

The system is called *Hamiltonian* if $\mathcal{D} = 0$, and *purely dissipative* if $\mathcal{H} = 0$.

A very useful representation of a quantum channel is needed for our derivation.

Kraus Operator Summation Representation (KOSR)

Any CP map $\Phi : \mathcal{B}(\mathscr{H}_1) \to \mathcal{B}(\mathscr{H}_2)$ can be decomposed into the form

Kraus rank, $r \leq \dim(\mathcal{B}(\mathscr{H}_1, \mathscr{H}_2))$

Kraus operators, $K_\alpha \in \mathcal{B}(\mathscr{H}_1, \mathscr{H}_2)$

$$\Phi(\rho) = \sum_{\alpha=1}^{r} K_\alpha \, \rho \, K_\alpha^\dagger \, . \tag{22}$$

The map is further TP iff it satisfies

$$\sum_{\alpha=1}^{r} K_\alpha^\dagger K_\alpha = \mathbb{1}_{\dim(\mathscr{H}_1)} . \tag{23}$$

Later on, we shall mainly focus on cases where $\mathscr{H}_2 = \mathscr{H}_1 \equiv \mathscr{H}_S$.

Proof: (KOSR)

For $\rho \in \mathcal{S}_\circ(\mathscr{H}_1)$, by Stinespring's dilation, there exists $\rho_E \in \mathcal{S}_\circ(\mathscr{H}_E)$ and isometry $U \in \mathcal{B}(\mathscr{H}_1 \otimes \mathscr{H}_E, \mathscr{H}_2 \otimes \mathscr{H}_E)$ such that

Since $\rho_E \in \mathcal{B}^\dagger(\mathscr{H}_E)$, we can decompose it with an orthonormal basis $\{\nu\}$ into $\rho_E = \sum_\nu \lambda_\nu |\nu\rangle\langle\nu|$.

$$\begin{aligned}
\Phi(\rho) &= \mathrm{Tr}_E\{U(\rho \otimes \rho_E)U^\dagger\} \\
&= \sum_\mu \langle\mu|U\rho \otimes \sum_\nu \lambda_\nu |\nu\rangle\langle\nu|U^\dagger|\mu\rangle \\
&= \sum_{\mu,\nu} \sqrt{\lambda_\nu}\langle\mu|U|\nu\rangle_E \rho \sqrt{\lambda_\nu}\langle\nu|U^\dagger|\mu\rangle_E \\
&= \sum_{\mu,\nu} K_{\mu\nu}\rho K_{\mu\nu}^\dagger = \sum_{\alpha=1}^{r} K_\alpha \rho K_\alpha^\dagger \qquad \blacksquare
\end{aligned}$$

Since $\nu, \mu \in \mathscr{H}_E$, $\langle\mu|U|\nu\rangle \in \mathcal{B}(\mathscr{H}_1, \mathscr{H}_2)$.

$\langle\nu|U^\dagger|\mu\rangle = (\langle\mu|U|\nu\rangle)^\dagger$

Proof: (Complete Positivity) For $A \succeq 0$,

$$\langle \nu | \Phi \otimes \mathbb{1}_d(A) | \nu \rangle = \sum_\alpha \langle \nu | (K_\alpha \otimes \mathbb{1}_d) A (K_\alpha^\dagger \otimes \mathbb{1}_d) | \nu \rangle = \sum_\alpha \langle w_\alpha | A | w_\alpha \rangle \geq 0 \qquad \blacksquare$$

Proof: (Trace-Preserving) For all ρ,

$$\mathrm{Tr}\left\{\Phi(\rho)\right\} = \sum_\alpha \mathrm{Tr}\left\{K_\alpha \rho K_\alpha^\dagger\right\} = \sum_\alpha \mathrm{Tr}\left\{K_\alpha^\dagger K_\alpha \rho\right\} = \mathrm{Tr}\left\{\sum_\alpha K_\alpha^\dagger K_\alpha \rho\right\}$$

$$\underbrace{\qquad\qquad}_{\text{Iff } \sum_\alpha K_\alpha^\dagger K_\alpha = \mathbb{1}_{\dim(\mathscr{H}_1)}} = \mathrm{Tr}\left\{\rho\right\} \qquad \blacksquare$$

A map has a Kraus operator summation representation iff it is linear and CPTP.

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Derivation (Short-Time Expansion)

Now, without further ado, let us give a quick derivation to the GKSL equation. The following proof shows that the GKSL equation is a generator of CPTP maps [Lid20, pp. 45–47].

Proof: (GKSL Equation)
Under the idea of short-time expansion, we Taylor expand $\rho(\mathrm{d}t)$ around the time $t = 0$, to obtain:

$$\rho(\mathrm{d}t) = \rho(0) + \dot{\rho}(0)\,\mathrm{d}t + o(\mathrm{d}t), \tag{24}$$

but for it to be a CPTP map, it must also coincide with a Kraus operator summation representation:

$$\rho(\mathrm{d}t) = \sum_{\alpha} K_\alpha(\mathrm{d}t)\rho(0)K_\alpha^\dagger(\mathrm{d}t).$$

Of what form can the Kraus operators be?

Derivation (Short-Time Expansion)

$$\begin{aligned}\rho(\mathrm{d}t) &= \rho(0) + \dot{\rho}(0)\,\mathrm{d}t + o(\mathrm{d}t)\\ &= \sum_{\alpha} K_\alpha(\mathrm{d}t)\rho(0)K_\alpha^\dagger(\mathrm{d}t)\end{aligned}$$

It can be seen that for

$$K_0 = \mathbb{1}_S + L_0\,\mathrm{d}t, \tag{25}$$

$L_0 \in \mathcal{B}(\mathscr{H}_S)$, we obtain the following:

$$\begin{aligned}K_0\rho(0)K_0^\dagger &= (\mathbb{1}_S + L_0\,\mathrm{d}t)\,\rho(0)\,(\mathbb{1}_S + L_0\,\mathrm{d}t)^\dagger\\ &= \rho(0) + \left(L_0\rho(0) + \rho(0)L_0^\dagger\right)\,\mathrm{d}t + o(\mathrm{d}t).\end{aligned}$$

There must be more K_α's.

Derivation (Short-Time Expansion)

The remaining Kraus operators must be of at least $O(\mathrm{d}t)$, hence the only possibility is

$$K_\alpha = \sqrt{\mathrm{d}t}\sqrt{\gamma'_\alpha}\, L_\alpha \qquad (\alpha = 1, \cdots, r-1). \tag{26}$$

unit: $1/t$ unitless, $\in \mathcal{B}(\mathscr{H}_S)$

The Kraus operators should also satisfy the TP condition:

$$\mathbb{1}_S = K_0^\dagger K_0 + \sum_{\alpha\geq 1} K_\alpha^\dagger K_\alpha = \mathbb{1}_S + \left(L_0 + L_0^\dagger + \sum_{\alpha\geq 1} |\gamma'_\alpha|\, L_\alpha^\dagger L_\alpha \right) \mathrm{d}t + o(\mathrm{d}t)$$

$$\Rightarrow L_0 + L_0^\dagger = -\sum_{\alpha\geq 1} \gamma_\alpha L_\alpha^\dagger L_\alpha.$$

$\gamma_\alpha := |\gamma'_\alpha| \geq 0$

Derivation (Short-Time Expansion)

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Hence, the quantum map will be

$$\rho(\mathrm{d}t) = \sum_{\alpha \geq 0} K_\alpha \rho(0) K_\alpha^\dagger$$

$$= \rho(0) + \left(L_0 \rho(0) + \rho(0) L_0^\dagger + \sum_{\alpha \geq 1} \gamma_\alpha L_\alpha \rho(0) L_\alpha^\dagger \right) \mathrm{d}t + o(\mathrm{d}t)$$

$$\Rightarrow \dot{\rho}(0) = L_0 \rho(0) + \rho(0) L_0^\dagger + \sum_{\alpha \geq 1} \gamma_\alpha L_\alpha \rho(0) L_\alpha^\dagger$$

$$L_0 = \frac{L_0 + L_0^\dagger}{2} + \frac{L_0 - L_0^\dagger}{2} =: -\frac{1}{2} \sum_{\alpha \geq 1} \gamma_\alpha L_\alpha^\dagger L_\alpha - i\mathcal{H}$$

Derivation (Short-Time Expansion)

$$\dot{\rho}(0) = -\frac{1}{2}\sum_{\alpha\geq 1}\gamma_\alpha L_\alpha^\dagger L_\alpha \rho(0) - i\mathcal{H}\rho(0)$$

$$-\frac{1}{2}\sum_{\alpha\geq 1}\gamma_\alpha \rho(0) L_\alpha^\dagger L_\alpha + i\rho(0)\mathcal{H} + \sum_{\alpha\geq 1}\gamma_\alpha L_\alpha \rho(0) L_\alpha^\dagger$$

$$= -i\left[\mathcal{H}, \rho(0)\right] + \sum_{\alpha\geq 1}\gamma_\alpha\left(L_\alpha\rho(0)L_\alpha^\dagger - \frac{1}{2}\{L_\alpha^\dagger L_\alpha, \rho(0)\}\right)$$

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\rho(t)\right|_{t=0} = \mathcal{L}\rho(0) \xrightarrow{\text{Markovian}} \frac{\mathrm{d}}{\mathrm{d}t}\rho(t) = \mathcal{L}\rho(t). \tag{27}$$

And thus we have obtained the GKSL equation in *Lindblad form*:

$$\dot{\rho}(t) = -i\left[\mathcal{H}, \rho(t)\right] + \sum_{\alpha\geq 1}\gamma_\alpha\left(L_\alpha\rho(t)L_\alpha^\dagger - \frac{1}{2}\{L_\alpha^\dagger L_\alpha, \rho(t)\}\right). \quad \blacksquare \tag{28}$$

Another form to the GKSL equation termed the *GKS form* can be obtained by a simple change of coordinates. Consider the Liouville space $\mathcal{B}(\mathscr{H})$ (with $N = \dim(\mathscr{H})$) with an ONB $\{F_i\}_{i=1}^{N^2}$ such that $\langle F_i, F_j \rangle = \mathrm{Tr}\left\{F_i^\dagger F_j\right\} = \delta_{ij}$. By decomposing the Lindblad operators as

$$L_\alpha = \sum_i C_\alpha^i F_i, \tag{29}$$

we have the GKSL equation as

$$\begin{aligned}\dot{\rho}(t) &= -i[\mathcal{H}, \rho(t)] + \sum_{\alpha \geq 1} \sum_{i,j=1}^{N^2} \gamma_\alpha C_\alpha^i \left(C_\alpha^j\right)^* \left(F_i \rho(t) F_j^\dagger - \frac{1}{2}\left\{F_j^\dagger F_i, \rho(t)\right\}\right) \\ &= -i[\mathcal{H}, \rho(t)] + \sum_{i,j=1}^{N^2} G_{ij} \left(F_i \rho(t) F_j^\dagger - \frac{1}{2}\left\{F_j^\dagger F_i, \rho(t)\right\}\right).\end{aligned}$$

Different Forms of the GKSL Equation

A recap on all the forms of the GKSL equation: for $\rho \in \mathcal{S}_\circ(\mathscr{H})$, $\mathcal{H} \in \mathcal{B}^\dagger(\mathscr{H})$,

1. Lindblad form:

$$\begin{aligned}\dot{\rho}(t) &= \sum_{\alpha\geq 1} K_\alpha \rho(t) K_\alpha^\dagger + L_0 \rho(t) + \rho(t) L_0^\dagger \\ &= -i[\mathcal{H}, \rho(t)] + \sum_{\alpha\geq 1} \gamma_\alpha \left(L_\alpha \rho(t) L_\alpha^\dagger - \frac{1}{2}\left\{ L_\alpha^\dagger L_\alpha, \rho(t) \right\} \right)\end{aligned}, \tag{30}$$

   where $\gamma_\alpha \geq 0$, $L_\alpha \in \mathcal{B}(\mathscr{H})$.

2. GKS form:

$$\dot{\rho}(t) = -i[\mathcal{H}, \rho(t)] + \sum_{ij} G_{ij} \left(F_i \rho(t) F_j^\dagger - \tfrac{1}{2}\left\{ F_j^\dagger F_i, \rho(t) \right\} \right), \tag{31}$$

   where the Kossakowski matrix $[G_{ij}] \succeq 0$, $\{F_i\}$ is an ONB of $\mathcal{B}(\mathscr{H})$.

The derivation above is a quick and simple one, utilizing the *Kraus operator summation representation* of our quantum map.

However, in order to gain more insights and to see when the GKSL equation is applicable to our quantum system, we shall gloss over a proof starting from *Stinespring's dilation*. Along the way, we will analyze the approximations used and when they are applicable.

Short-Time Expansion

Stinespring's Dilation

$$\rho(t) = \mathrm{Tr}_E \left\{ U \rho_T(0) U^\dagger \right\}$$

*Assume separable initial state $\rho_T(0) = \rho(0) \otimes \rho_E$

Kraus Operators Summation Representation

$$\rho(t) = \sum_{\alpha=1}^{r} K_\alpha \rho(0) K_\alpha^\dagger$$

Compare with Taylor expansion. Find Ansatz for Kraus operators.

$$K_0 = \mathbb{1} + L_0 \, \mathrm{d}t$$

$$K_\alpha = \sqrt{\mathrm{d}t} \sqrt{\gamma_\alpha'} \, L_\alpha$$

$$L_0 = -\frac{1}{2} \sum_{\alpha \geq 1} \gamma_\alpha L_\alpha^\dagger L_\alpha - i\mathcal{H}$$

Assume Markovian

$$\dot{\rho}(0) = \mathcal{L}\rho(0) \rightarrow \rho(t) = \mathrm{e}^{t\mathcal{L}} \rho(0)$$

Lindblad Form

$$\dot{\rho}(t) = -i[\mathcal{H}, \rho(t)] + \sum_{\alpha \geq 1} \gamma_\alpha \left(L_\alpha \rho(t) L_\alpha^\dagger - \frac{1}{2} \left\{ L_\alpha^\dagger L_\alpha, \rho(t) \right\} \right)$$

Change of Coordinates

GKS Form

$$\dot{\rho}(t) = -i[\mathcal{H}, \rho(t)] + \sum_{i,j} G_{ij} \left(F_i \rho(t) F_j^\dagger - \frac{1}{2} \left\{ F_j^\dagger F_i, \rho(t) \right\} \right)$$

(By following the derivation on the handouts,) The approximations used tell us when the GKSL equation is applicable [BP07, pp. 130–136]:

- **Weak coupling (Born approximation)**:
  The interaction between the system and the environment is small, hence we can expand to $o(\hat{\mathcal{H}}_I^2)$. Moreover, the environment is only weakly affected by the interaction, hence $\hat{\rho}_T(t) = \hat{\rho}(t) \otimes \hat{\rho}_E$. Furthermore, we can even assume the environment to be stationary.
- **Markov approximation**:
  The excitation time scale of environment $\tau_E \ll \tau_S$, the time scale over which the system varies considerably. In other words, the correlation function $\mathrm{Tr}\left\{\hat{E}_j(s)\hat{E}_i(0)\hat{\rho}_E\right\}$ decays sufficiently fast compared to τ_S.

Remarks on GKSL Equation

Note that the equation also works in infinite dimensions, as is in [Lin76].

During the derivation, we implicitly implied $\mathcal{L}$ is bounded and assumed *a certain degree* of continuity.

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Heisenberg's Picture

In the Heisenberg's picture, the linear operators evolve according to the semigroup generated by the adjoint infinitesimal generator $\mathcal{L}^\dagger$.

$$\mathcal{L}^\dagger A = i\left[\mathcal{H}, A\right] + \sum_\alpha \gamma_\alpha \left(L_\alpha^\dagger A L_\alpha - \frac{1}{2}\left\{ L_\alpha^\dagger L_\alpha, A \right\} \right). \tag{32}$$

We can see that $\mathcal{L}^\dagger(\mathbb{1}) = 0$, hence it is indeed a generator of TP channels.

For reference:

$$\mathcal{L}\rho = -i\left[\mathcal{H}, \rho\right] + \sum_\alpha \gamma_\alpha \left(L_\alpha \rho L_\alpha^\dagger - \frac{1}{2}\left\{ L_\alpha^\dagger L_\alpha, \rho \right\} \right).$$

The Lindblad master equation is invariant under:

1. unitary transformation of Lindblad operators:

$$\sqrt{\gamma'_\alpha} L_\alpha \mapsto \sqrt{\gamma''_\alpha} L'_\alpha = \sum_\alpha u_{\alpha'\alpha} \sqrt{\gamma'_\alpha} L_\alpha .$$

2. homogeneous transformations of the form:

$$\begin{aligned} L_\alpha &\mapsto L'_\alpha = L_\alpha + \kappa_\alpha \mathbb{1} \\ \mathcal{H} &\mapsto \mathcal{H}' = \mathcal{H} + \frac{1}{2i} \sum_{\alpha \geq 0} \gamma_\alpha \left(\kappa^*_\alpha L_\alpha - \kappa_\alpha L^\dagger_\alpha \right) + \kappa \mathbb{1}, \end{aligned} \tag{33}$$

where $\kappa \in \mathbb{R}$, $\kappa_\alpha \in \mathbb{C}$.

Henceforth, we can always find traceless L_α's without loss of generality.

Purity Reduction

Under the Lindblad form of dynamics, if all the Lindblad operators L_α are Hermitian, i.e. $L_\alpha \in \mathcal{B}^\dagger(\mathscr{H})$, then the purity of the system fulfills $\frac{\mathrm{d}}{\mathrm{d}t}\mathsf{Tr}\left\{\rho^2\right\} \leq 0$.

Proof:

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathsf{Tr}\left\{\rho^2\right\} = 2\left(\sum_\alpha \gamma_\alpha \mathsf{Tr}\left\{\rho L_\alpha \rho L_\alpha^\dagger\right\} - \sum_\alpha \gamma_\alpha \mathsf{Tr}\left\{\rho^2 L_\alpha^\dagger L_\alpha\right\}\right)$$

$$\overset{M_\alpha = L_\alpha \rho}{=} -2\sum_\alpha \gamma_\alpha \underbrace{\mathsf{Tr}\left\{\left(M_\alpha^\dagger - M_\alpha\right) M_\alpha\right\}}_{\geq 0} \leq 0$$

The last inequality uses the Schur decomposition to obtain: for any sqaure matrix M, $||M||_F \geq ||M||_2$. ■

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Conclusion

- A quantum channel has to be linear and CPTP.
- A time-independent Markovian channel must be of the form $\mathrm{e}^{\mathcal{L}t}$.
- A quantum channel can be represented using Stinespring's dialtion and Kraus operators:

$$\Phi(\rho) = \mathrm{Tr}_E \left\{ U(\rho \otimes \rho_E) U^\dagger \right\} = \sum_\alpha K_\alpha \rho K_\alpha^\dagger, \; U\text{: isometry}, \; \sum_\alpha K_\alpha^\dagger K_\alpha = \mathbb{1}.$$

- The Lindbladian is a generator of Markovian CPTP maps, it is of the form

$$\dot{\rho} = \mathcal{L}\rho = -i\left[\mathcal{H}, \rho\right] + \sum_\alpha \gamma_\alpha \left(L_\alpha \rho L_\alpha^\dagger - \frac{1}{2}\left\{ L_\alpha^\dagger L_\alpha, \rho \right\} \right),$$

applicable in weak-coupling system with fast decaying excitement in the environment.

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For [GKS76; Kos72; Lin76], not much were touched on. The derivations included are much too mathematical for me to understand. For detailed derivation to the GKSL equation, also refer to [BP07; Lid20]. For divisibility, refer to [WC08]. [Led23] contains detailed analysis of quantum channels. And lastly, [CP17] includes a short walk-through of the history.